

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc (Physics) Examination May 2018

Course code & Name PS 955: Nuclear reactor, Nuclear Energy & Nuclear Accelerators

Date: 4th May 2018 Day: Friday Time: 10.00 To 10.20

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	10
1.	The interaction time in compound nucleus is _____ times nuclear transit time a) 10^3 b) 10^4 c) 10^5 d) 10^6	
2.	The total macroscopic cross-section of a material is 0.25 cm^{-1} , then its mean free path will be: a) 4 cm b) 8 cm c) 16 cm d) 24 cm	
3.	In (n, γ) reaction cross-section varies a) inversely with velocity b) square of velocity c) linearly with velocity 4 cm d) Independent of velocity	

4.	In an exothermic reaction the reaction to take one needs projectile energy to be a) Greater than threshold energy b) Much greater than threshold energy c) Zero d) Equal to threshold energy	
5.	For a reactivity of 40×10^{-4} , the effective neutron multiplication factor is a) 1.002 b) 1.004 c) 1.006 d) 1.008	
6.	For heavy nuclei at glancing angle the energy loss in single collision will be a) 100 % b) 50% c) No loss. d) 25%	
7.	One radiation dose absorbed is roughly equal to a) 10 ergs/gm. b) 100 ergs/gm. c) 1 ergs/gm. d) 1000 ergs/gm.	
8.	Roentgen equivalent Man (REM) is a) ratio of dose in RAD by RBE b) RAD x RBE c) RBE/RAD d) None of the above	
9.	LINAC produces well-collimated beam compare to cyclic accelerators. (TRUE/FALSE)	
10.	Cyclotron has wide radial width of the RF cavity, while microtone has zero RF cavity (TRUE/FALSE)	

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Course code & Name PS955 Nuclear reactor, Nuclear energy & Nuclear accelerators

Date: 4th May 2018 Day: Friday Time: 10.20 To 12.00 Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed		10
a	Define the term Microscopic and macroscopic cross-sections.	02
b	How direct and compound nucleus differs	02
c	Write the full form of RAD, RBE and REM and relation between these terms.	02
d	Why cyclotron has to be modified for producing high energy particles?	02
e	Define the terms: Slowing down power and Moderating ratio.	02
SECTION - II		
Q-III Answer the following questions as directed		15
a	Calculate the mean-free path and the macroscopic cross section of Zr of atomic weight 91.22 gm and whose density is 6.5 gm/cm ³ and microscopic absorption and scattering cross-sections are 0.18 and 8 barns, respectively	3
b	Calculate the threshold energy for $^{19}\text{F} (n, p)^{19}\text{O}$ $m_{\text{F}} = 18.998404 \text{ amu}$, $m_{\text{p}} = 1.007825 \text{ amu}$, $m_{\text{O}} = 19.0035774 \text{ amu}$ $m_{\text{n}} = 1.008665 \text{ amu}$	3
c	Match the following for reaction $a + X = b + Y$ 1) For (n , n) reaction 1) Cross-section σ varies inversely with velocity 2) For (p, γ) reaction 2) σ varies as $(1/v) e^{-Ga}$ 3) For (n, γ) reaction 3) Q is negative 4) For (n, n') reaction 4) cross section is independent of energy	3
d	Explain the working of Betatron and obtain the Betatron condition.	3
e	Write four factor formula for neutron cycle and explain each terms.	3